

EXP NO: 04
DATE:
06/08/2026

INVERTING AND NON-INVERTING AMPLIFIER

INVERTING AMPLIFIER

Aim:

To design an inverting amplifier using op-amp for the given gain of 2.

Apparatus Required

S.NO	Components	Specification
1	Operational Amplifier	IC 741
2	Resistors	1K- Ω , 2K- Ω
3	Breadboard / Trainer Kit	1
4	CRO (Cathode Ray Oscilloscope)	1
5	Function Generator	1
6	Connecting wires	As required

Design

$$A_v = -R_f/R_{in}$$

$$A_v = -2$$

$$-2 = -R_f/R_{in}$$

$$R_{in} = 1K\Omega$$

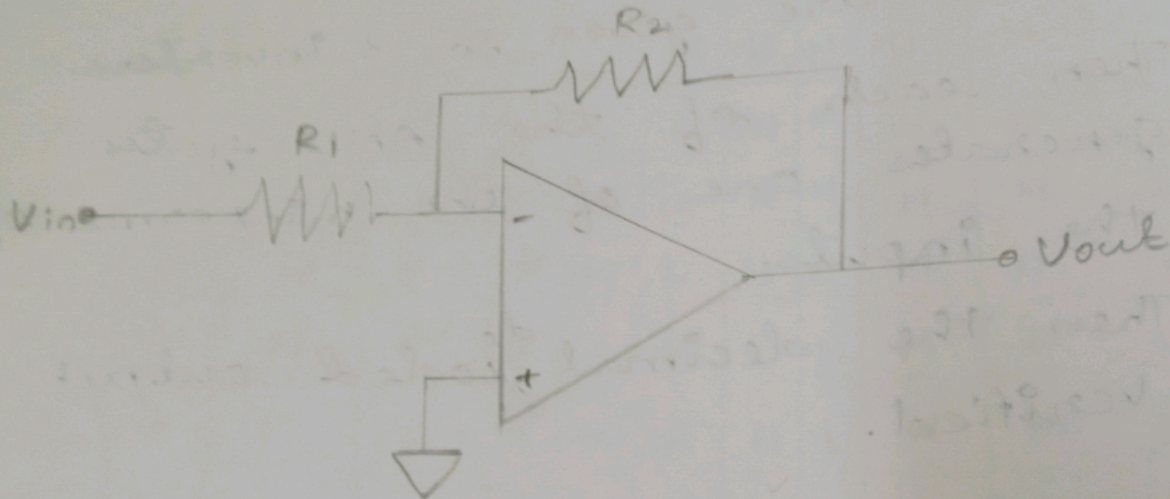
$$R_f = 2K\Omega$$

Output
equation

$$V_o = A_v V_{in}$$

$$V_o = -\frac{R_f}{R_{in}} V_{in}$$

Inverting Amplifier circuit



OBSERVATION TABLE

Sr No	Vin (V)	Vout (V)	Practical Gain	Theoretical Gain
1	1	-2.6	-2.6	-2.2
2	2	-5	-2.5	-2.2
3	3	-7.2	-2.4	-2.2
4	4	-9.4	-2.35	-2.2

Procedure

1. Connect IC 741 on the trainer kit.
2. Give dual power supply (+VCC and -VCC) to the IC.
3. Make the circuit connections as per the circuit diagram.
4. Apply the sinusoidal input signal from the function generator to the inverting terminal.
5. Ground the non-inverting terminal.
6. Connect feedback resistor (R-f) between output and inverting terminal.
7. Switch on the power supply and CRO.
8. Observe input and output waveforms on the CRO.
9. Verify that the output waveform is amplified and 180° out of phase with the input.
10. Tabulate the readings.

Result:

The working of the inverting amplifier was studied and verified successfully.

NON-INVERTING AMPLIFIER.

Aim:-

To design a non-inverting amplifier using an op-amp for the given gain of 2.

Apparatus Required:-

S.NO	Components	Specifications
1	Operational Amplifier	IC 741
2	Resistors	1K- Ω , 2K- Ω
3	Function Generator	1
4	CRO	1
5	Dual Power supply	± 12 V
6	Trainer Kit	1
7	Connecting wires	As required

Design

For a non-inverting Amplifier

$$V_o = \left(1 + \frac{R_f}{R_i} \right) V_{in}$$

Gain

$$A_v = 1 + \frac{R_f}{R_i}$$

$$A_v = 2$$

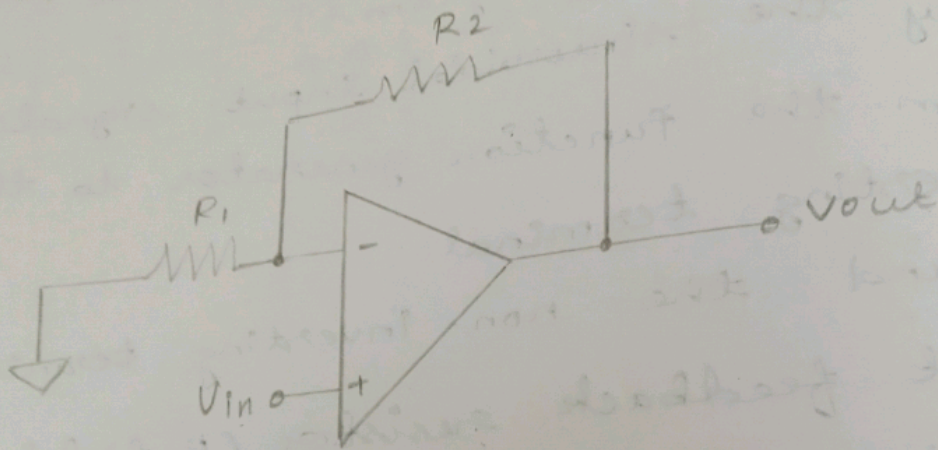
$$2 = 1 + R_f / R_i$$

$$R_f = R_i$$

$$R_i = 1K-\Omega$$

$$R_f = 1K-\Omega$$

Non-Inverting Amplifier circuit



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OBSERVATION TABLE

S.NO	V _{in} (V)	V _{out} (V)	Practical Gain	Theoretical Gain
1	0.5V	1.38	3.76	2.2
2	1V	2.94V	3.94	2.2
3	1.5V	4.73V	4.153	2.2
4	2V	6.26V	4.13	2.2
5	2.5V	7.96V	4.18	2.2
6	3V	9.50V	4.16	2.2
7	3.6V	10.57V	3.926	2.2

Procedure

1. Place IC 741 on the trainer kit.
2. Connect dual power supply $\pm 12V$ to the IC.
3. Make connections as per the circuit diagram.
4. Apply sinusoidal input signal from function generator to the non-inverting terminal.
5. Connect feedback resistor and resistor properly.
6. Switch on the power supply and CRO.
7. Observe input and output waveforms on the CRO.
8. Measure input voltage, output voltage, time period and frequency.
9. Verify that the output waveform is amplified and in phase with input waveform.
10. Tabulate the readings.

Result :-

The non-inverting amplifier was designed and verified successfully.